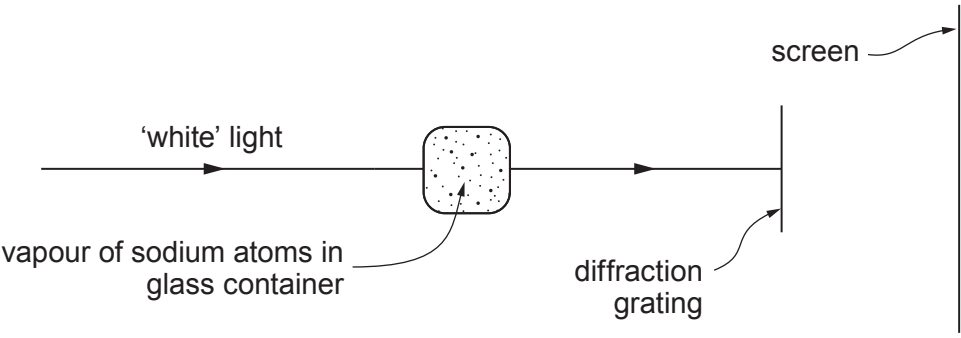


6. (a) The diagram shows how the *absorption spectrum* of sodium atoms may be produced.



Describe the appearance of the absorption spectrum, comparing it with the *emission spectrum* of sodium atoms. [3]

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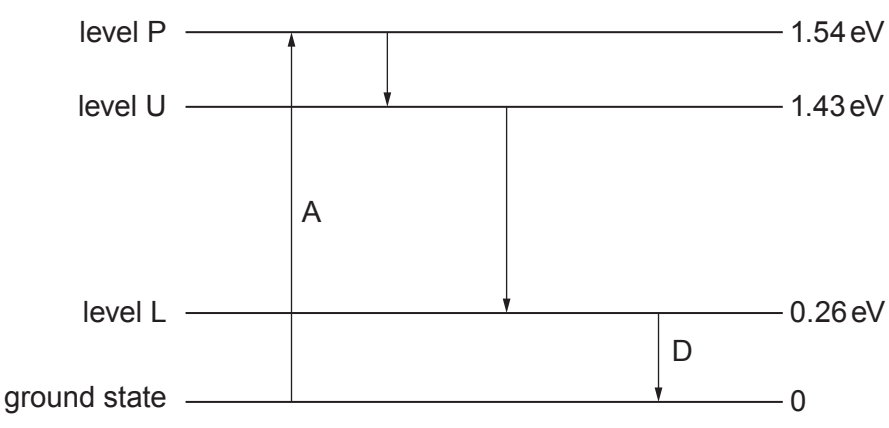
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(b) A simplified energy level diagram for the amplifying medium of a 4 level laser is given.



Examiner
only

(i) Referring to *populations*, explain the part played in the operation of the laser by:

I. transition A; [2]

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II. transition D. [2]

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(ii) Calculate the wavelength of the radiation emitted by stimulated emission from the laser, and name the region of the electromagnetic spectrum in which it lies. [4]

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(c) Most laser pointers produce polarised light. Discuss whether or not students attending a lecture in which a laser pointer is used should be given spectacles fitted with polarising filters (polaroids) to wear for safety during the lecture. [3]

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